

Lecture 4: Classical Logic and Quantifiers

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1 Lecture 4: Classical Logic and Quantifiers

1.1 Aim of the lecture

Lecture 3 connected Lean tactics with the introduction and elimination rules of natural deduction. This lecture extends that framework in two directions. We first introduce *reductio ad absurdum* as a classical rule. We then move from propositional logic to predicate logic and learn to represent objects, properties, functions, relations, and quantifiers.

The lecture will:

- distinguish constructive rules from classical principles;
- apply *reductio ad absurdum*;
- build a small first-order language in Lean;
- formalize statements containing $\forall$ and $\exists$;
- identify and apply $\forall I$, $\forall E$, $\exists I$, and $\exists E$.

1.2 From constructive rules to classical logic

The rules seen so far have a direct constructive reading: each proof specifies how to build the required term. To prove a conjunction we construct both components; to prove an implication we construct a function; to prove an existential statement, as we will see, we provide a witness.

In Lean's constructive base logic, excluded middle $A \vee \neg A$ and double-negation elimination $\neg\neg A \rightarrow A$ are not automatically available. Lean does allow classical principles to be used locally and explicitly (Source: [Theorem Proving in Lean 4, Classical Logic](#)).

1.3 Reductio ad absurdum

Classical *reductio ad absurdum* proves A by showing that assuming $\neg A$ leads to `False`. In Lean, we apply it with `Classical.byContradiction`.

$$\begin{array}{c}
 \frac{}{\neg A} \quad 1 \\
 \vdots \\
 \frac{\perp}{A} \quad 1 \quad \text{RAA}
 \end{array}$$

Reductio ad absurdum, RAA

The working pattern is: apply reductio, introduce $\neg A$, derive **False** using the available assumptions, and use exactly that contradiction.

```

theorem lecture04_reductio_ad_absurdum
  (Rains TakeUmbrella : Prop)
  (hNotRainsNotUmbrella : ¬Rains → ¬TakeUmbrella)
  (hTakeUmbrella : TakeUmbrella) :
  Rains := by
  apply Classical.byContradiction
  intro hNotRains
  have hNotTakeUmbrella := hNotRainsNotUmbrella hNotRains
  have hContradiction := hNotTakeUmbrella hTakeUmbrella
  exact hContradiction

```

The first assumption turns $\neg$ Rains into $\neg$ TakeUmbrella. The latter is a function from TakeUmbrella to **False**; applying it to hTakeUmbrella produces the required contradiction.

1.4 A limitation of propositional logic

In propositional logic, each complete statement is represented by a separate proposition. We might use P for “Hypatia is a philosopher,” Q for “Hannah Arendt is a philosopher,” and R for “Hypatia is curious.” This hides the fact that P and Q attribute the same property to different objects. It also prevents us from directly expressing “every philosopher is curious” or “there is a curious philosopher.”

Predicate logic exposes the objects we discuss and the properties or relations attributed to them.

1.5 The domain of discourse

Every formalization begins with a *domain*: the collection of objects under consideration. In this lecture we use one general type.

```

variable (Thing : Type)

```

Thing represents the domain of discourse. A term $x : \text{Thing}$ is an element of that domain. This is the usual single-domain presentation of first-order logic (Source: [Logic and Proof, First Order Logic in Lean](#)).

Named objects are represented by terms of the domain:

```
variable (Thing : Type)
variable (hypatia hannah rome athens : Thing)
```

These declarations do not yet assert any proposition. They only introduce terms that will receive an interpretation.

1.6 Predicates: properties of objects

A one-place predicate takes an object and returns a proposition.

```
variable (Thing : Type)
variable (Person City Philosopher Curious : Thing → Prop)
variable (hypatia hannah : Thing)

#check Philosopher hypatia
#check Philosopher hannah
```

Philosopher hypatia and Philosopher hannah are different propositions, but they share the same predicate.

Natural language	Lean
Hypatia is curious	Curious hypatia
Hannah Arendt is curious	Curious hannah
Io is a moon of Jupiter	MoonOfJupiter io
It rains in Rome	RainsIn rome

1.7 Functions: constructing new terms

A function takes one or more objects from the domain and returns an object.

```
variable (Thing : Type)
variable (hypatia : Thing)
variable (motherOf : Thing → Thing)
variable (Curious : Thing → Prop)

#check motherOf hypatia
#check Curious (motherOf hypatia)
```

motherOf hypatia : Thing is a new term. Since its type is still Thing, it can be used as the argument of a predicate. Functions can be nested: motherOf (motherOf hypatia) denotes the mother of Hypatia's mother.

In first-order logic a function is total and returns exactly one result for each argument. A function returns an object; a predicate returns a proposition.

1.8 Relations between objects

A predicate with two or more arguments expresses a relation.

```
variable (Thing : Type)
variable (hypatia hannah athens : Thing)
variable (Knows Visits : Thing → Thing → Prop)

#check Knows hypatia hannah
#check Visits hypatia athens
```

Argument order matters. `Knows hypatia hannah` and `Knows hannah hypatia` represent different statements.

1.9 Local parameters and assumptions

The keyword `variable` introduces local parameters. A section limits their scope, and Lean adds to a theorem only the parameters actually used in its statement.

```
section PredicateLogicLanguage

variable (Thing : Type)
variable (hypatia hannah : Thing)
variable (Philosopher Curious : Thing → Prop)
variable (motherOf : Thing → Thing)
variable (Knows : Thing → Thing → Prop)

end PredicateLogicLanguage
```

Declaring `Philosopher : Thing → Prop` does not assert that anyone is a philosopher. A genuine assumption that Hypatia is a philosopher has the form `hHypatiaPhilosopher : Philosopher hypatia`. The vocabulary determines which propositions can be formulated; assumptions provide proofs of some of them.

1.10 Variables and quantifiers

A variable `x : Thing` represents an object that has not been specified. Quantifiers turn expressions containing such variables into statements:

- $\forall$ reads “for every”;
- $\exists$ reads “there exists at least one.”

Natural language	Lean
Every person is curious	$\forall x : \text{Thing}, \text{Person } x \rightarrow \text{Curious } x$
Some person is curious	$\exists x : \text{Thing}, \text{Person } x \wedge \text{Curious } x$
Every philosopher is curious	$\forall x : \text{Thing}, \text{Philosopher } x \rightarrow \text{Curious } x$
There is a curious philosopher	$\exists x : \text{Thing}, \text{Philosopher } x \wedge \text{Curious } x$

Writing $x : \text{Thing}$ explicitly makes both the introduced variable and its domain visible.

1.11 Implication and conjunction under quantifiers

Two patterns occur repeatedly:

```
variable (Thing : Type)
variable (Philosopher Curious : Thing → Prop)

#check ∀ x : Thing, Philosopher x → Curious x
#check ∃ x : Thing, Philosopher x ∧ Curious x
```

In the universal formula we take an arbitrary thing and say: *if* it is a philosopher, then it is curious. The implication restricts the assertion to philosophers. In the existential formula we seek one witness that is *both* a philosopher *and* curious, so the properties are joined by a conjunction.

```
variable (Thing : Type)
variable (Student Reads Understands : Thing → Prop)

#check ∀ x : Thing, Student x → Reads x
#check ∃ x : Thing, Student x ∧ Reads x
#check ∀ x : Thing, Student x → Reads x → Understands x
#check ∃ x : Thing, Student x ∧ Reads x ∧ Understands x
```

#check verifies that an expression is a well-formed proposition. It does not prove that proposition.

1.12 Functions inside formulas

Terms built with functions can occur inside predicates, relations, and quantified statements.

```
variable (Thing : Type)
variable (hypatia : Thing)
variable (Student Reads Curious : Thing → Prop)
variable (motherOf : Thing → Thing)
variable (Knows : Thing → Thing → Prop)

#check Curious (motherOf hypatia)
#check ∀ x : Thing, Student x → Knows x (motherOf x)
#check ∃ x : Thing, Student x ∧ Reads (motherOf x)
```

The parentheses show that we first build `motherOf x` and then use that term as the argument of a predicate or relation.

1.13 Nested quantifiers

Quantifier order determines which choices may depend on others.

```

variable (Thing : Type)
variable (Person : Thing → Prop)
variable (Knows : Thing → Thing → Prop)

-- Every person knows someone.
#check ∀ x : Thing, Person x →
  ∃ y : Thing, Person y ∧ Knows x y

-- Someone knows every person.
#check ∃ x : Thing, Person x ∧
  ∀ y : Thing, Person y → Knows x y

-- Nobody knows every person.
#check ∃¬ x : Thing, Person x ∧
  ∀ y : Thing, Person y → Knows x y

```

In the first statement, the witness y may change when x changes. The second statement requires one fixed x who knows every relevant y .

1.14 Universal introduction, ‘ $\forall$ I’

$$\frac{A(x)}{\forall y A(y)} \forall I$$

Universal quantifier introduction, $\forall I$

To prove $\forall y : \text{Thing}, A y$, introduce an arbitrary x . The goal becomes $A x$. Prove it without using any special property of x ; the conclusion then holds for every object in the domain.

```

theorem lecture04_forall_intro
  (Thing : Type)
  (Person Curious : Thing → Prop) :
  ∀ y : Thing, Person y → Curious y → Curious y := by
  intro x
  intro hXPerson
  intro hXCurious
  exact hXCurious

```

Only the first `intro` applies $\forall I$. The following two introduce the assumptions of the implications.

1.14.1 The arbitrariness condition

The object introduced by $\forall I$ cannot be a particular object selected because of a special property. In Lean, `intro x` introduces a fresh object of the type over which we are quantifying.

1.15 Universal elimination, ‘ $\forall E$ ’

$$\frac{\forall x A(x)}{A(t)} \forall E$$

Universal quantifier elimination, $\forall E$

Given $\forall x : \text{Thing}, A x$, choose a term t and apply the universal proof to it. This yields $A t$.

```
theorem lecture04_forall_elim
  (Thing : Type)
  (Student Reads : Thing → Prop)
  (marta : Thing)
  (hEveryStudentReads : ∀ x : Thing, Student x → Reads x)
  (hMartaStudent : Student marta) :
  Reads marta := by
  have hIfMartaStudentThenReads := hEveryStudentReads marta
  have hMartaReads := hIfMartaStudentThenReads hMartaStudent
  exact hMartaReads
```

Applying the universal assumption to `marta` produces `Student marta → Reads marta`, which can then be applied to `hMartaStudent`.

1.16 Existential introduction, ‘ $\exists I$ ’

$$\frac{A(t)}{\exists x A(x)} \exists I$$

Existential quantifier introduction, $\exists I$

To prove $\exists x : \text{Thing}, A x$, choose a witness t and prove $A t$. Lean’s constructor is `Exists.intro`.

```
theorem lecture04_exists_intro
  (Thing : Type)
  (Reads : Thing → Prop)
  (marta : Thing)
  (hMartaReads : Reads marta) :
  ∃ x : Thing, Reads x := by
  apply Exists.intro marta
  exact hMartaReads
```

Choosing `marta` turns the existential goal into `Reads marta`, which is already available as an assumption.

1.17 Existential elimination, ‘ $\exists E$ ’

$$\frac{\exists x A(x) \quad \begin{array}{c} \overline{A(y)}^1 \quad 1 \\ \vdots \\ B \end{array}}{B} \exists E$$

Existential quantifier elimination, $\exists E$

Given $\exists x : \text{Thing}$, $A\ x$, we know that a witness exists but may not assume its identity. To derive B , introduce an arbitrary witness y together with $A\ y$, and derive a conclusion that does not depend on the identity of y .

```
theorem lecture04_exists_elim
  (Thing : Type)
  (FiledReport : Thing → Prop)
  (InvestigationStarts : Prop)
  (hSomeoneFiledReport : ∃ x : Thing, FiledReport x)
  (hReportStartsInvestigation :
    ∀ x : Thing, FiledReport x → InvestigationStarts) :
  InvestigationStarts := by
  apply Exists.elim hSomeoneFiledReport
  intro y
  intro hYFiledReport
  have hInvestigation :=
    hReportStartsInvestigation y hYFiledReport
  exact hInvestigation
```

The conclusion `InvestigationStarts` contains no occurrence of `y` and does not depend on the witness’s identity.

1.18 Combining the rules

The quantifier rules are often used together.

```
theorem lecture04_exists_elim_combined
  (Thing : Type)
  (HasUmbrella StaysDry : Thing → Prop)
  (hSomeoneHasUmbrella : ∃ x : Thing, HasUmbrella x)
  (hUmbrellaStaysDry : ∀ x : Thing, HasUmbrella x → StaysDry x) :
  ∃ x : Thing, StaysDry x := by
  apply Exists.elim hSomeoneHasUmbrella
  intro x
  intro hXHasUmbrella
  apply Exists.intro x
  have hXStaysDry := hUmbrellaStaysDry x hXHasUmbrella
  exact hXStaysDry
```


Here `Exists.elim` opens the first existential, applying `hUmbrellaStaysDry` to `x` uses $\forall E$, and `Exists.intro` `x` uses the same object as the witness of the new existential.

1.19 Working strategy

When reading a quantified formula, ask:

- Does the goal begin with $\forall$? Introduce an arbitrary object with `intro`.
- Is there a universal assumption? Apply it to a term from the domain.
- Does the goal begin with $\exists$? Choose a witness with `Exists.intro`.
- Is there an existential assumption? Open it with `Exists.elim` without presupposing the witness's identity.
- Are several quantifiers nested? Check their order and dependencies.

This extends the strategy used for propositional connectives: inspect the main constructor of the goal for an introduction rule and inspect compound assumptions for useful elimination rules.

1.20 Sources and further reading

- Jeremy Avigad, Robert Y. Lewis, and Floris van Doorn, *Logic and Proof*, “Natural Deduction Rules”: https://leanprover.github.io/logic_and_proof_lean3/nd_quickref.html.
- Lean developers, *The Lean Language Reference*, “Quantifiers”: <https://lean-lang.org/doc/reference/latest/Basic-Propositions/Quantifiers/>.
- Jeremy Avigad, Robert Y. Lewis, and Floris van Doorn, *Logic and Proof*, “First Order Logic in Lean”: https://leanprover.github.io/logic_and_proof_lean3/first_order_logic_in_lean.html.
- Jeremy Avigad, Leonardo de Moura, Soonho Kong, and Sebastian Ullrich, *Theorem Proving in Lean 4*, “Classical Logic”: https://lean-lang.org/theorem_proving_in_lean4/Propositions-and-Proofs/.